Homework 1

Problem 1. Rossby Wave Instability (50 points)

Consider a two-dimensional, inviscid, non-magnetic disk in cylindrical coordinates (r, ϕ). Neglect disk self-gravity. Let Σ denote surface density and P vertically integrated pressure. Adopt the barotropic equation of state $P = c_s^2\Sigma$, with constant c_s. The governing equations are

$$\frac{\partial \Sigma}{\partial t} + \nabla \cdot (\Sigma \boldsymbol{v}) = 0, \tag{1}$$

$$\begin{aligned} \frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v} \cdot \nabla)\boldsymbol{v} &= -\frac{\nabla P}{\Sigma} - \nabla\Phi, \\ \Phi &= -\frac{GM_*}{r}. \end{aligned} \tag{2}$$

The axisymmetric equilibrium has $v_{r0} = 0$ and $v_{\phi 0} = r\Omega(r)$, with

$$\Sigma_0(r) = \Sigma_{\rm b}\left[1 + A\exp\left(-\frac{(r-r_0)^2}{2w^2}\right)\right]. \tag{3}$$

Here $A > 0$ and $w \ll r_0$. Define

$$\begin{aligned} \kappa^2 &= \frac{1}{r^3}\frac{d(r^4\Omega^2)}{dr}, \\ \zeta_0 &= \frac{\kappa^2}{2\Omega\Sigma_0}. \end{aligned} \tag{4}$$

Consider non-axisymmetric perturbations proportional to $e^{i(m\phi - \omega t)}$, with integer $m \geq 1$. Define the Doppler-shifted frequency $\widetilde{\omega}(r) = \omega - m\Omega(r)$ and the enthalpy perturbation $W(r) = \delta P/\Sigma_0$.

1. **Equilibrium and linearization (15 points).** Derive $\Omega^2(r)$ from radial force balance and write the linearized continuity, radial-momentum, and azimuthal-momentum equations. State the conditions needed for $\Omega^2 > 0$ and $\kappa^2 > 0$.

2. **Radial eigenvalue equation (20 points).** Eliminate δv_r, δv_ϕ, and $\delta\Sigma$ to obtain a second-order ordinary differential equation for $W(r)$. Write every coefficient explicitly in terms of background quantities and ω. Show where the vortensity ζ_0 gradient enters the equation and identify the corotation radius $r_{\rm c}$, defined by $\mathrm{Re}(\omega) = m\Omega(r_{\rm c})$.

3. **Boundary conditions and instability (10 points).** Consider a finite domain $r_{\rm in} < r < r_{\rm out}$ enclosing the bump, with impermeable boundaries $\delta v_r = 0$. Translate these conditions into conditions on W. Explain why ω is an eigenvalue and give the growth condition under the stated time convention. Distinguish the role of an extremum of the vortensity ζ_0 from a sufficient condition for an unstable global mode.

4. **Physical interpretation (5 points).** Explain how Rossby waves on opposite sides of an extremum of the vortensity ζ_0 can interact. Explain why a density bump alone does not justify assuming that the disk is unstable.

Problem 2. Magnetorotational Instability (50 points)

Use a local Cartesian frame (x, y, z) rotating at angular frequency Ω, with x, y, and z pointing in the radial, azimuthal, and vertical directions. Adopt an inviscid, incompressible ideal-MHD model. The reference density ρ_0 is uniform in space and constant in time:

$$\rho_0 = \text{constant}.$$

$$\delta\rho = 0.$$

Neglect vertical gravity and take the Keplerian shear parameter $q = 3/2$. The governing equations are

$$\nabla \cdot \boldsymbol{v} = 0,$$

$$\nabla \cdot \boldsymbol{B} = 0,$$

$$\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v} \cdot \nabla)\boldsymbol{v} = -\frac{\nabla P}{\rho_0} - 2\Omega \boldsymbol{e}_z \times \boldsymbol{v} + 2q\Omega^2 x \boldsymbol{e}_x + \frac{(\nabla \times \boldsymbol{B}) \times \boldsymbol{B}}{4\pi\rho_0}, \tag{5}$$

$$\frac{\partial \boldsymbol{B}}{\partial t} = \nabla \times (\boldsymbol{v} \times \boldsymbol{B}).$$

Take the background flow, pressure, and magnetic field to be

$$\begin{aligned} \boldsymbol{v}_0 &= -q\Omega x \boldsymbol{e}_y, \\ P_0 &= \text{constant}, \\ \boldsymbol{B}_0 &= B_0 \boldsymbol{e}_z, \\ B_0 &= \text{constant}. \end{aligned} \tag{6}$$

Write $\boldsymbol{v} = \boldsymbol{v}_0 + \delta\boldsymbol{v}$, $\boldsymbol{B} = \boldsymbol{B}_0 + \delta\boldsymbol{B}$, and $P = P_0 + \delta P$. Consider axisymmetric perturbations proportional to $e^{\omega t + ikz}$, with vertical wavenumber $k > 0$. Here ω is the complex growth rate and $v_A = B_0/\sqrt{4\pi\rho_0}$ is the background Alfvén speed.

1. **Equilibrium and constraints (10 points).** Verify that the prescribed background satisfies the steady momentum and induction equations. Use incompressibility and the solenoidal magnetic-field constraint to determine δv_z and δB_z.

2. **Linearized equations (15 points).** Derive the radial and azimuthal momentum and induction equations for δv_x, δv_y, δB_x, and δB_y. Retain Coriolis forces, background shear, and magnetic tension. Explain why the pressure perturbation does not enter the horizontal momentum equations for the specified perturbations. Write the resulting four-variable system in matrix form.

3. **Dispersion relation (15 points).** Set the determinant of the system to zero and derive the MRI dispersion relation as a polynomial in ω. Write all coefficients explicitly in terms of Ω, k, and v_A. Verify that the limit $B_0 \to 0$ recovers the hydrodynamic epicyclic oscillation branch.

4. **Growth and physical interpretation (10 points).** Determine the range of kv_A/Ω for which a growing MRI mode exists. Find the maximum growth rate and the corresponding wavenumber. Explain how magnetic tension transfers angular momentum between radially displaced fluid elements and drives the instability.

Suggested reading.

Lovelace et al. (1999), ApJ, 513, 805. DOI
Li et al. (2000), ApJ, 533, 1023. DOI
Balbus & Hawley (1991), ApJ, 376, 214. DOI